Artificial Intelligence-Introduction

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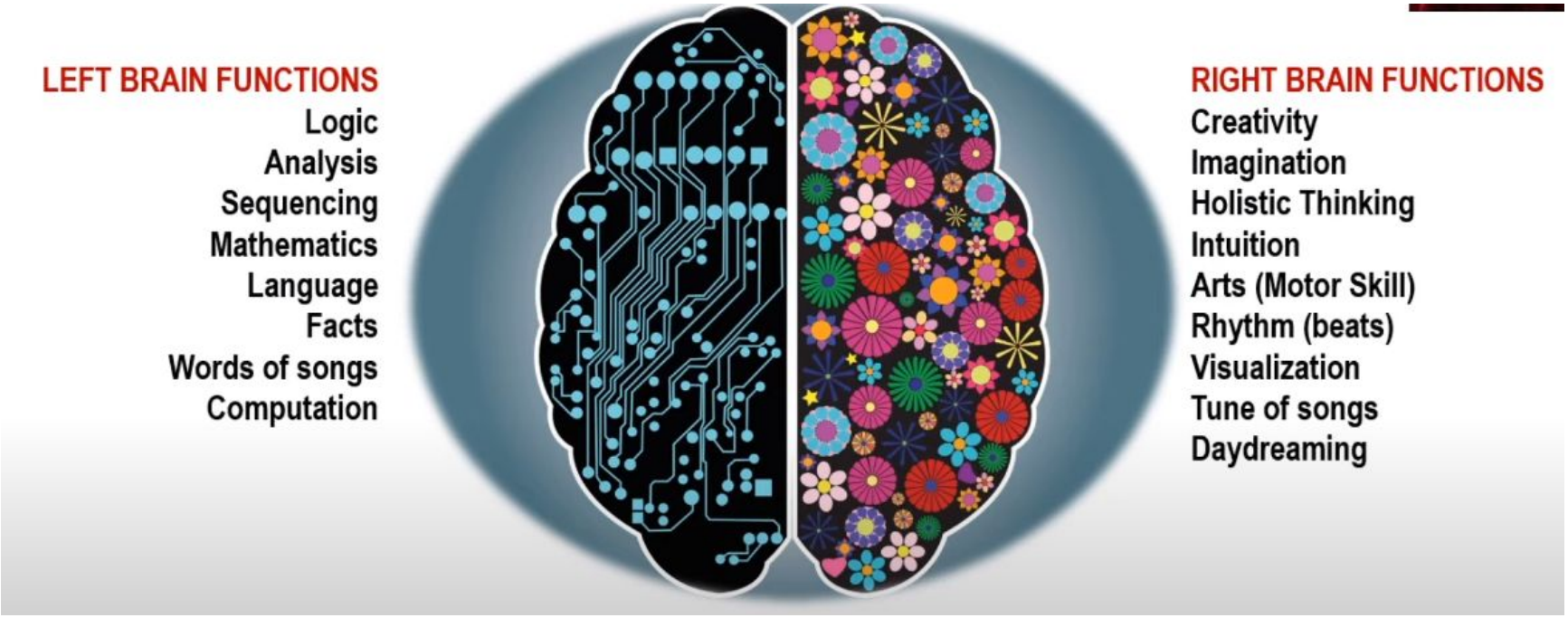
Cognitive System of Human



Classical AI

Modern AI: Convergence of these two

Statistical Methods, Machine Learning

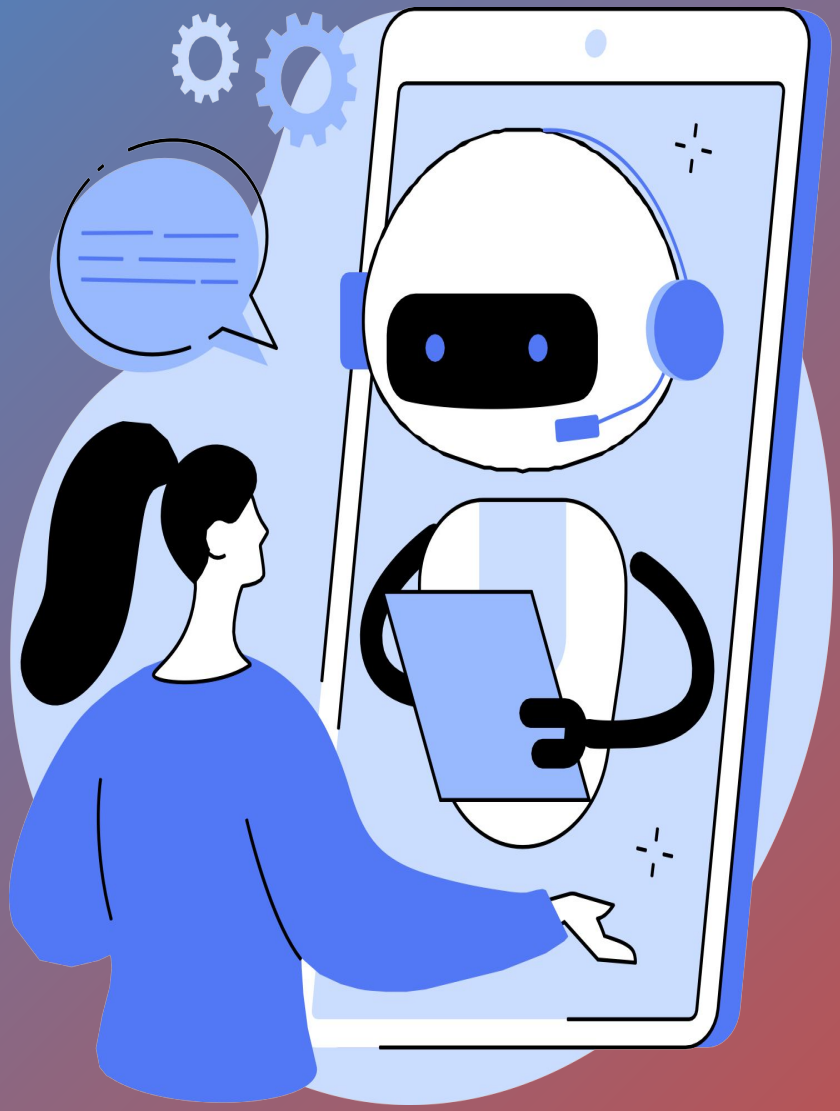
Stuart Russell, Peter Norvig, *Artificial intelligence : A Modern Approach*, Prentice Hall, Fourth edition, 2020.

Motivations

- AI is not just a technical subject, but it has relation with Philosophy.
- AI subject consisting of many sub-fields.
- AI based problems have many solutions.
- Breadth wise course not depth (Something of Everything)
- Tradeoff between Broad and Shallow vs. Deep and Narrow.
- Searching, heuristic search, planning, decision making, Reinforcement Learning.
- Theory → Modeling → Algorithm → Applications

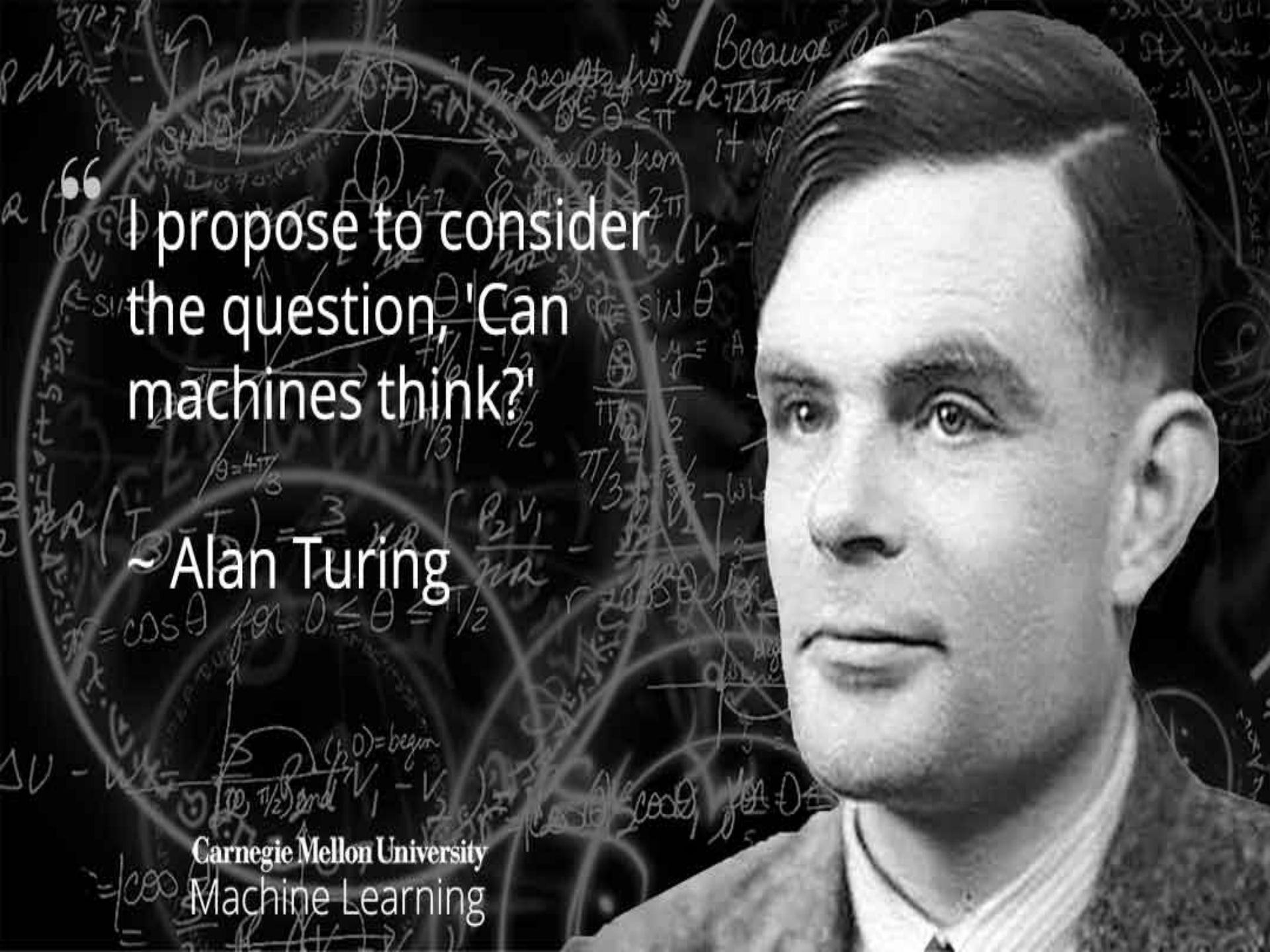
Motivations

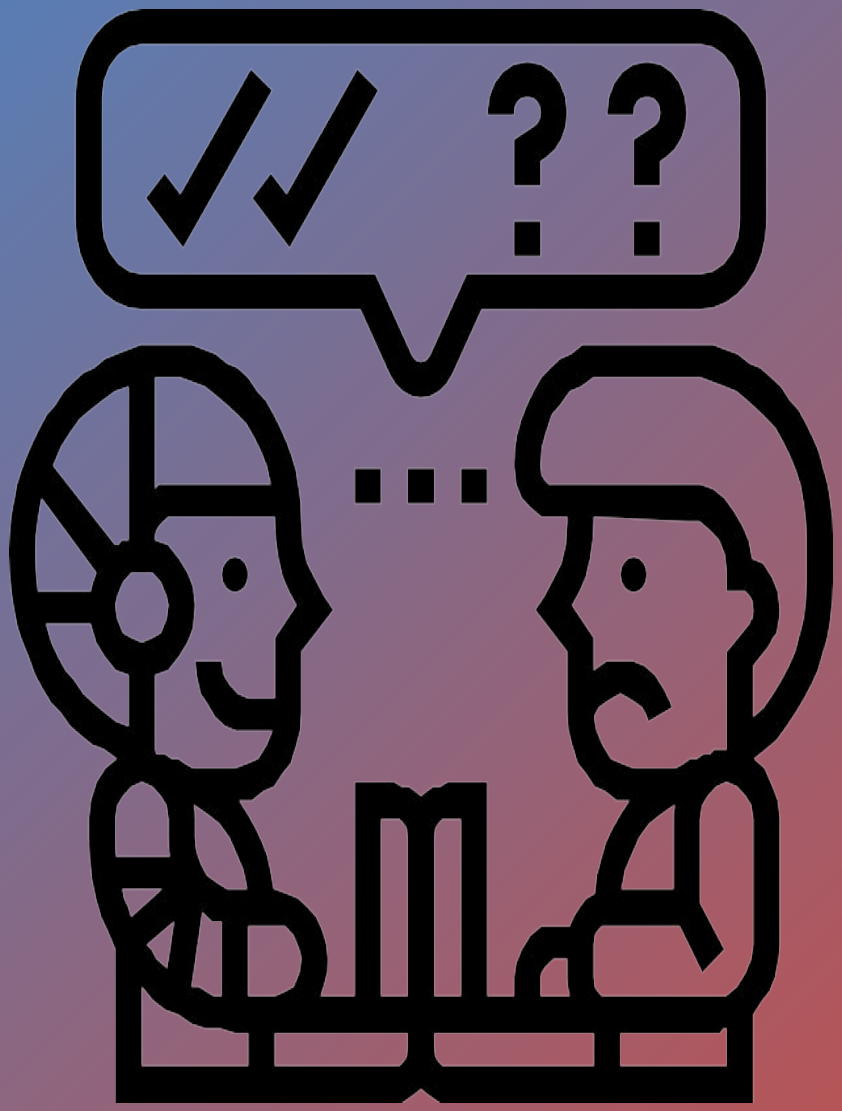
- Take a real life problem, convert it into a computational problem, i.e. Modeling
- For finding solution, apply suitable algorithms for the problem.
- ENIAC is the first computer (1946) following principle of Turing Machine, used for US defense purposes.
- Lot of progress of AI has done due to Defense
- Question: “Can a machine think?” – Alan Turing in 1950
- Not known if the machine cannot communicate with us by any way.
- Brain is meaningless in absence of communicators.



Can Machines Think?

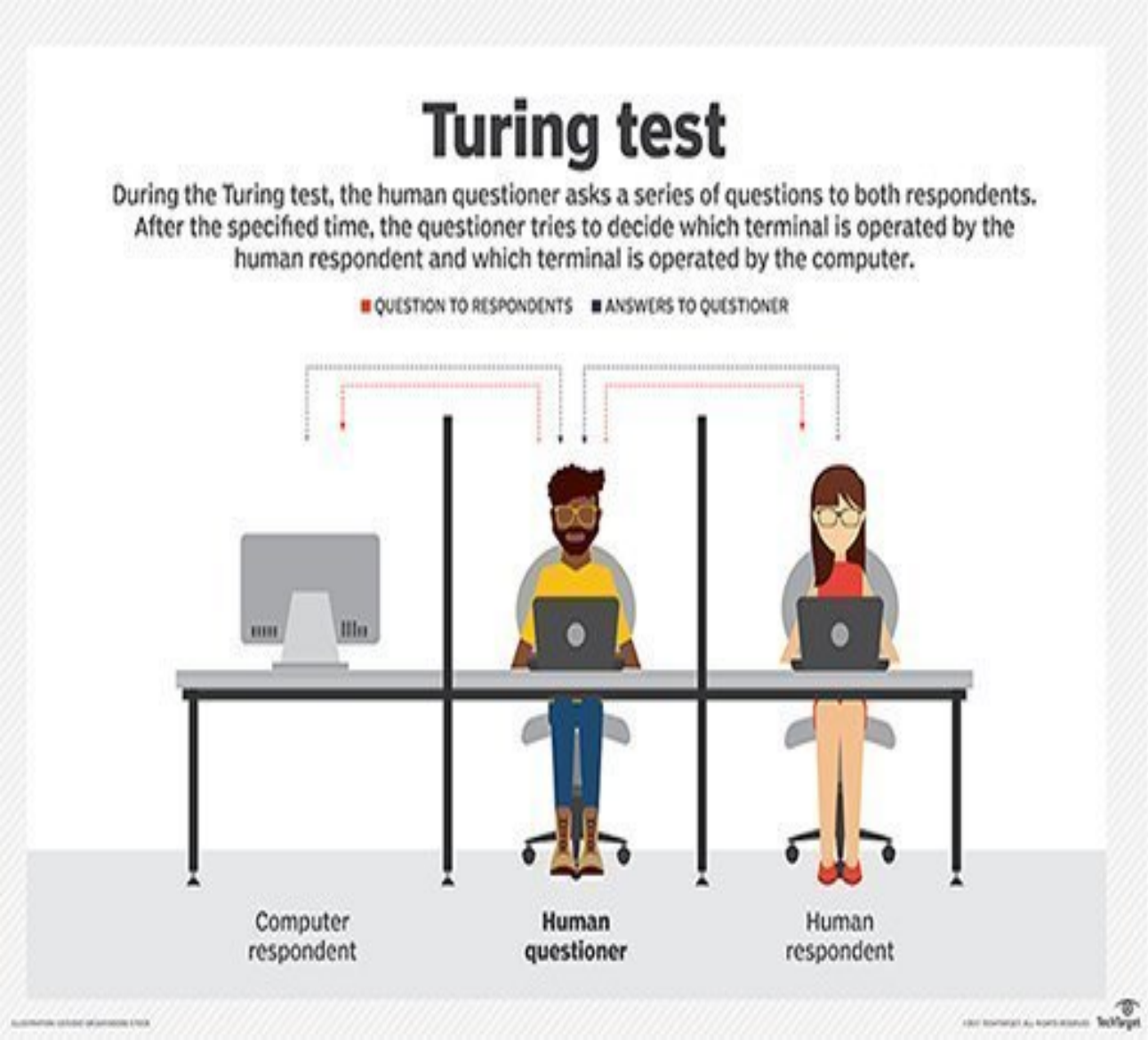
- Ever since the possibility of building intelligent machines arose, there have been raging debates on whether machine intelligence is possible or not.
- All kinds of arguments have been put forth both for and against the possibility. It was perhaps to put an end to these arguments that Alan Turing (1950) proposed his famous imitation game, which we now now call the Turing Test.


“I propose to consider the question, 'Can machines think?'
~ Alan Turing
Carnegie Mellon University
Machine Learning



Experimental Setup and Findings of the Turing Test

- The Turing Test, proposed by Alan Turing in 1950, is a method to evaluate a machine's ability to exhibit intelligent behavior indistinguishable from that of a human. The setup involves three participants:
  - **Human Interrogator (Judge):** A person who poses questions to both a human and a machine.
  - **Human Participant:** A person who responds to the interrogator's questions.
  - **Machine:** A computer or AI program designed to generate responses.
- The interrogator communicates with both the human and the machine through a text- based interface, ensuring that physical presence or voice cues do not give away the participants' identities.
- The test typically involves a series of conversations where the interrogator asks various questions and tries to distinguish between the human and the machine.


Turing test
During the Turing test, the human questioner asks a series of questions to both respondents. After the specified time, the questioner tries to decide which terminal is operated by the human respondent and which terminal is operated by the computer.
QUESTION TO RESPONDENTS
ANSWERS TO QUESTIONER
Computer respondent
Human questioner
Human respondent

- 1950: Turing Test for Machine Intelligence

- 1956: AI born at Dartmouth College Wrkshop

McDermott

- 1964: Eliza – the chatbot psychotherapist

1966: Shakey – general purpose mobile robot

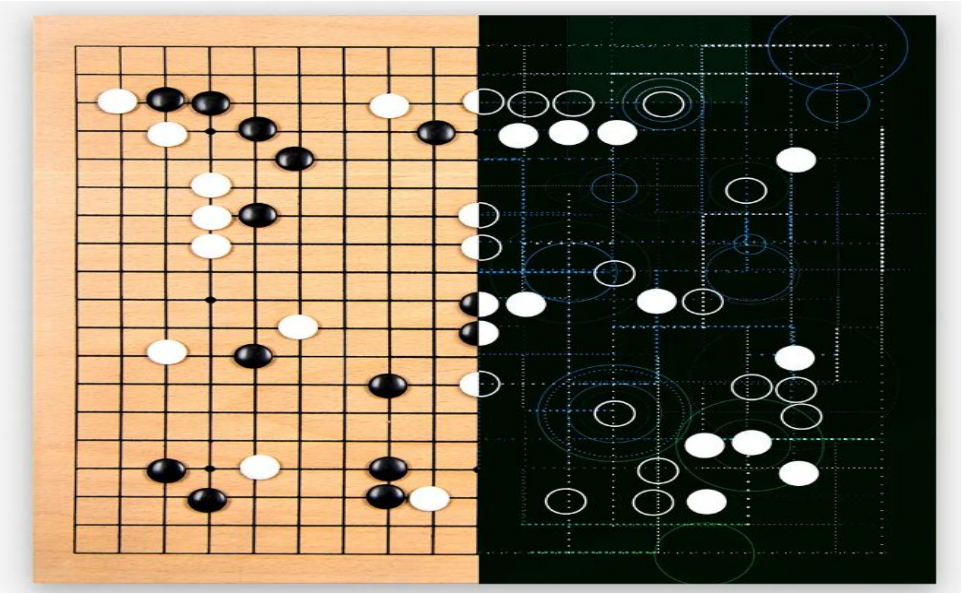
1974-1980 no development on AI

Next Expert systems were developed but not sustained.

1997 – AI ends human supremacy in Chess

Phase of AI

- First Phase – Logic based AI
- Second Phase – Probabilistic AI
- Third phase – Neural Network based AI
- AlphaGo competition
  - **Go** (game) is an abstract strategy board game for two **players** in which the aim is to surround more territory than the opponent.



- Go---a 2,500-year-old game that's exponentially more complex than chess.
- Researchers at DeepMind---a self-professed "Apollo program for AI" that Google acquired in 2014---staged this machine-versus-man contest in October, in London.
- IN A MAJOR breakthrough for artificial intelligence, a computing system developed by Google researchers in Great Britain has beaten a top human player at the game of Go.

What is Artificial Intelligence

- Artificial Intelligence (AI) is the use of computer systems to simulate human thought processes, for performing tasks that typically require human intelligence, such as learning, problem-solving, and decision-making.

- Artificial Intelligence (AI) is the branch of computer science that creates intelligent machines which act like humans

- It is one of the revolutionizing technologies that people are fascinated for its ability to relate to their daily lives

- AI enables machines enhance and automate tasks across industries

Tasks: Artificial Intelligence

- Ability to perform tasks normally requiring human intelligence, such as **visual** perception, **speech** recognition, **decision-making**, and **translation** between languages
- Ability of a computer program or a machine to **think** and **learn**
- Ability to correctly interpret external data, to learn from such data, and to use those learnings to achieve specific goals and tasks through flexible adaptation
- Ability to mimic human **cognition**
- A program that can **sense**, **reason**, **act** and **adapt**

Capabilities of Artificial Intelligence

- Natural Language Processing
- Recognizing Patterns
- Making Decisions
- Learning from Experience

Definitions of AI

Father of AI: John McCarthy

Defined AI as *"The science and engineering of making intelligent machines".*

Bellman

"The automation of activities associated with human thinking, like decision-making, problem-solving."

1956 — **1959** — **1978**

Samuel's Self-Learning Program

"Developed the first self-learning program, the checkers-playing AI, illustrating that machines could learn from experience."

Definitions of AI

Haugeland
"The exciting new effort to make computers think... machines with minds".

Rich and Knight
"Making computers do things at which, at the moment, people are better".

Winston
"The study of computations that make it possible to perceive, reason, and act".

1985 1991

1985 1991 1992

Charniak and McDermott
"The study of mental faculties through computational models".

Partridge
"A collection of algorithms that are adequate approximations of intractable problems".

Features of Artificial Intelligence

Ability to Learn

AI systems can improve their performance eventually by learning from data and past experiences.

Logical Decision Making

AI systems are fed with large amounts of data to understand and recognize patterns for analysis and decision making.

Adaptability

AI systems can adjust and adapt to changes in data.

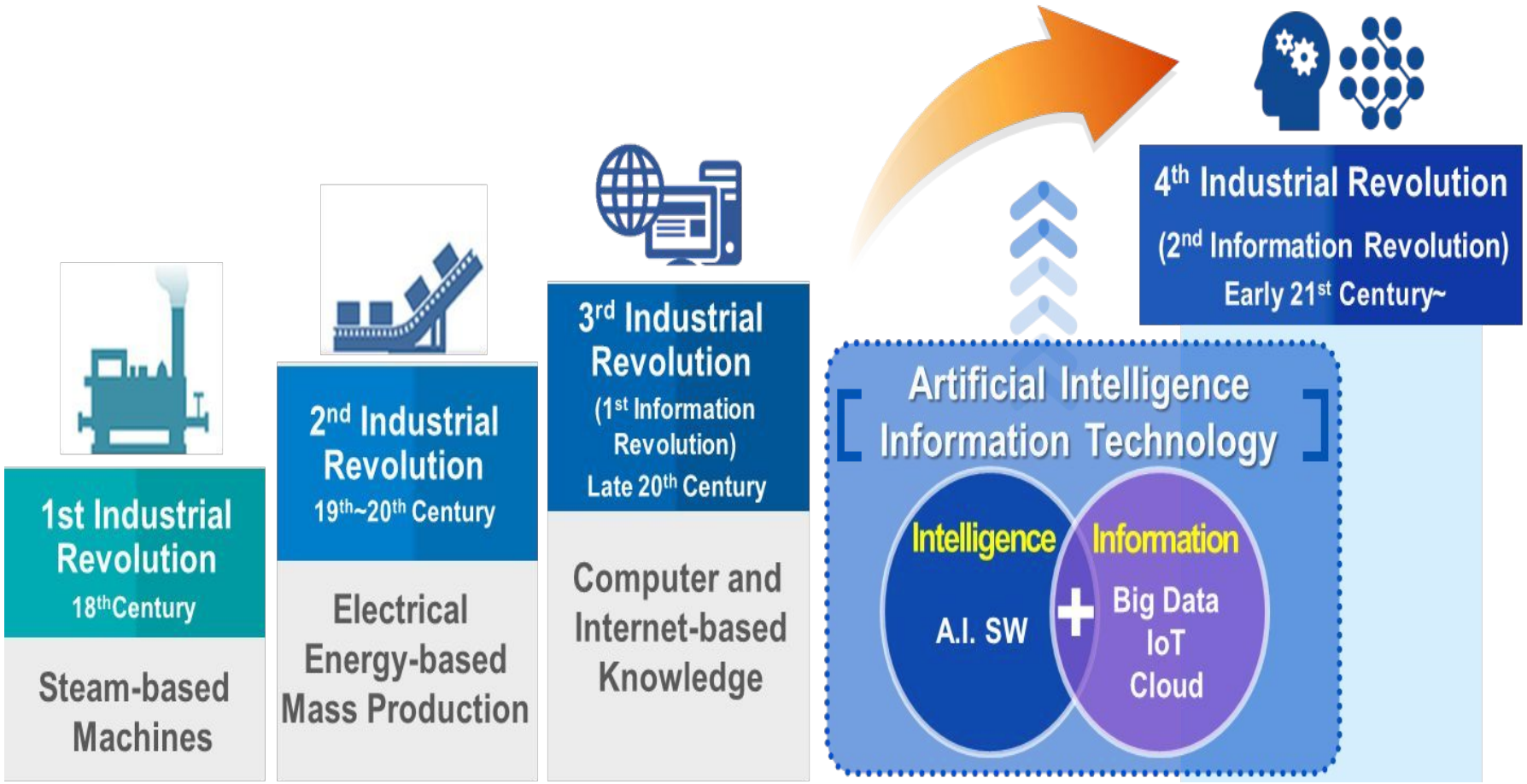
Efficient Automation

AI would efficiently execute repetitive tasks and processes.

Versatility

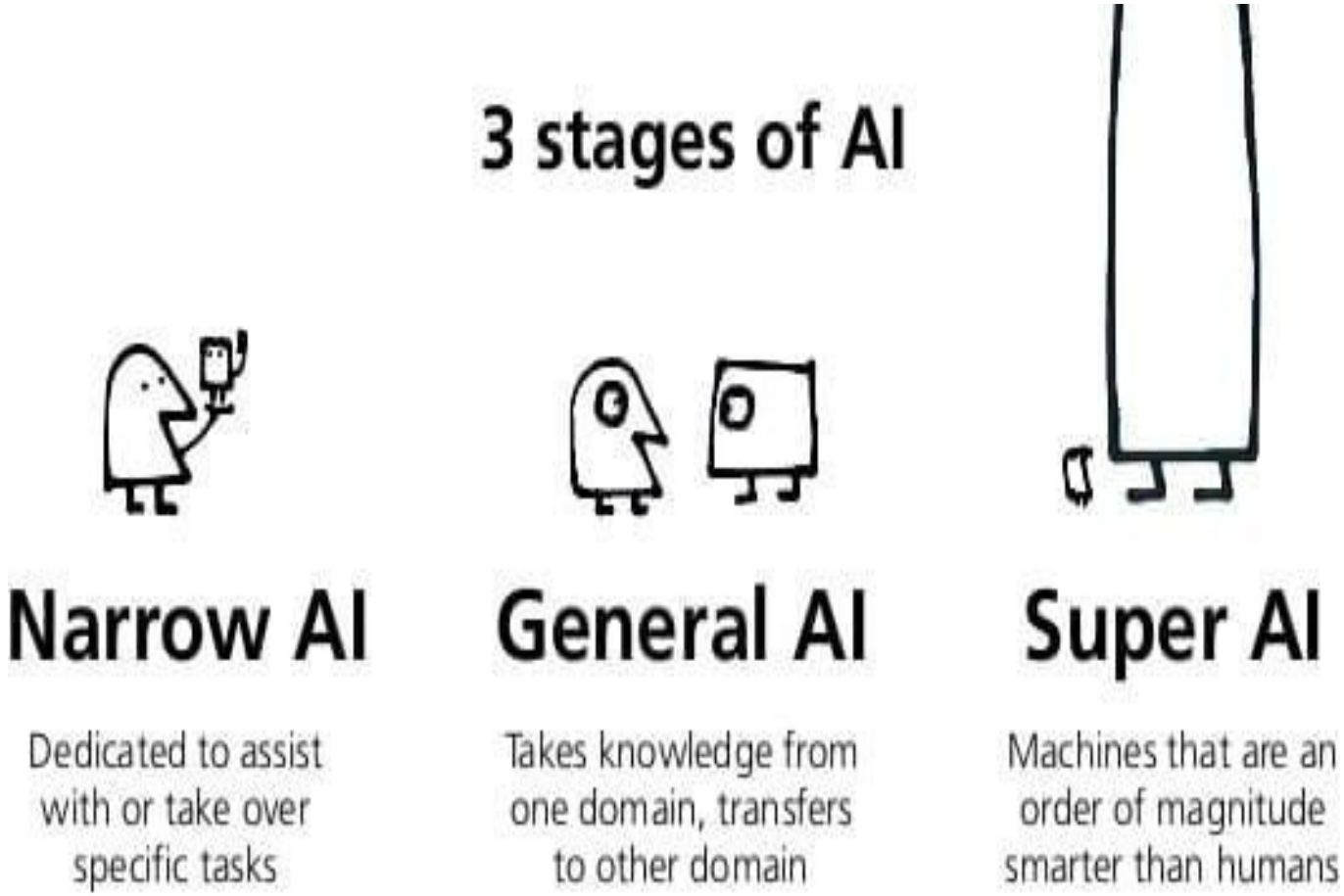
AI can be widely applied for various tasks across all fields like businesses, automotive, health, and many others.

Evolution of Industry



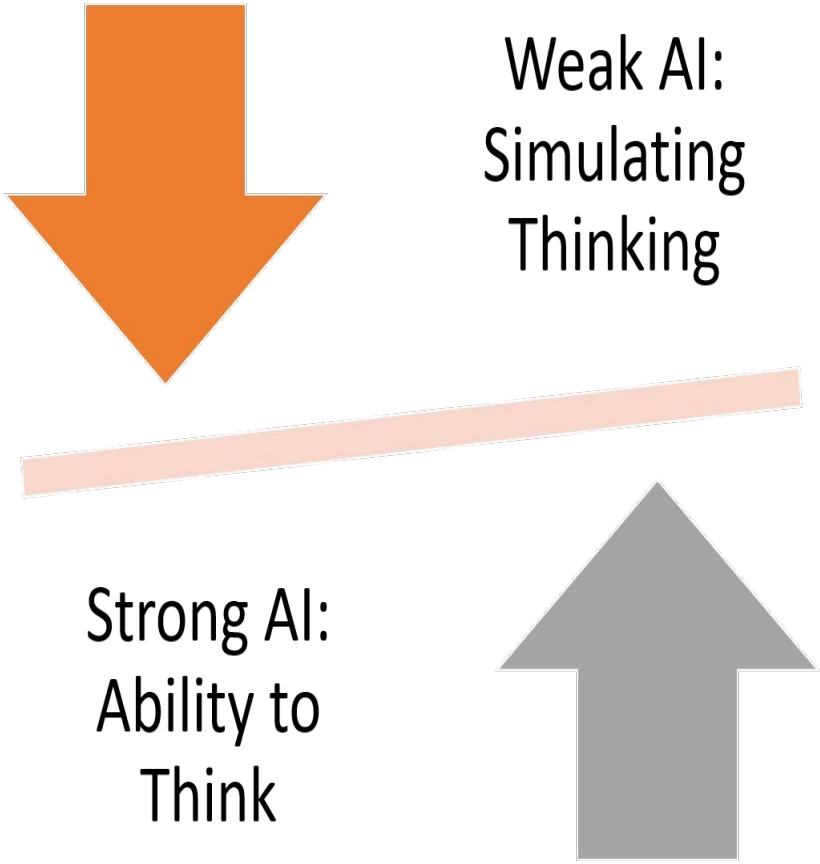
Source:
https://blogs.worldbank.org/digital-development/what-korea-s-strategy-manage-implications-artificial-intelligence

3 Stages of AI



Source: [AI & Data Preparation – Avoiding 'Garbage-In, Garbage-Out']

Weak AI vs Strong AI



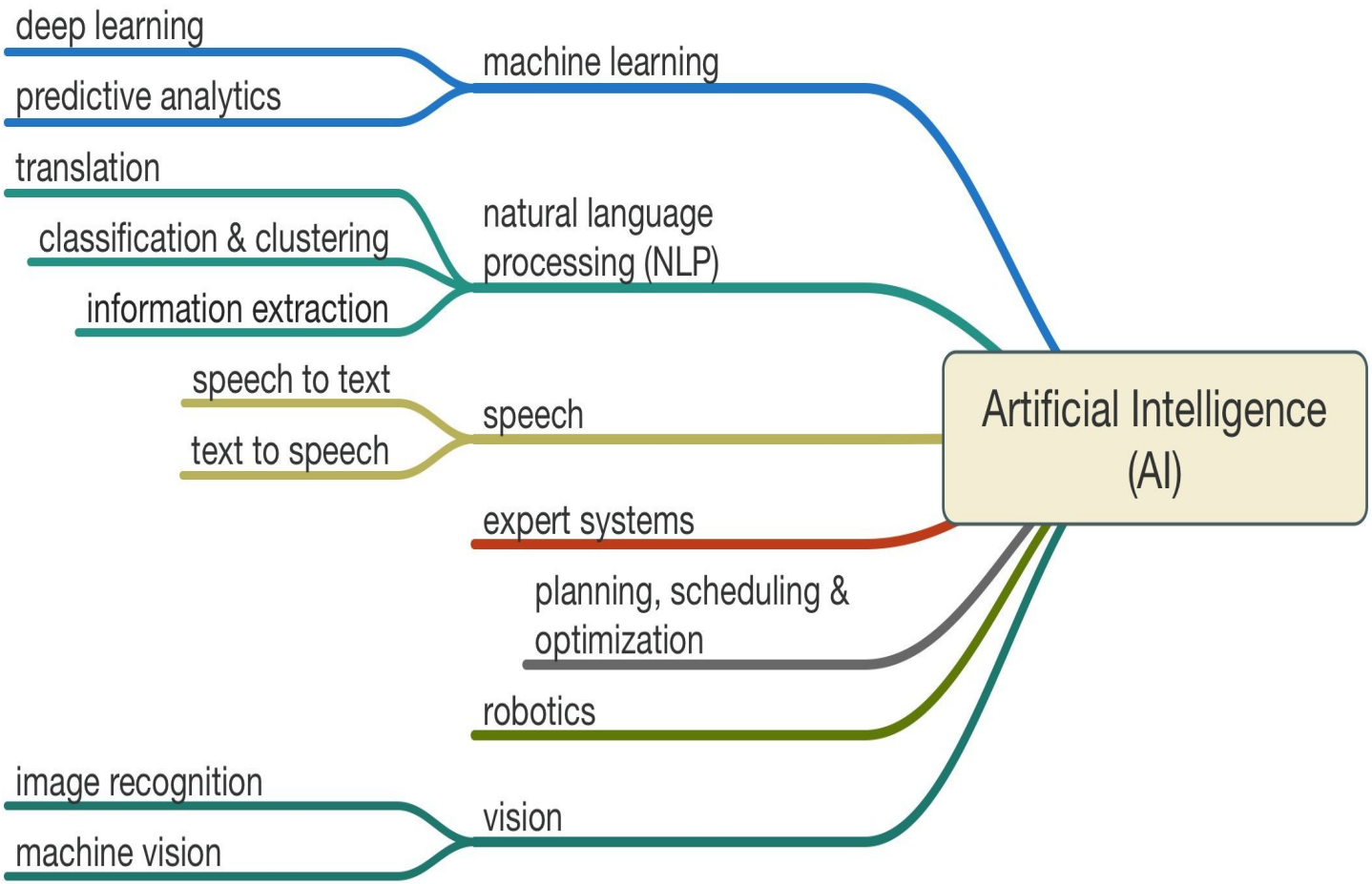
Weak AI vs Strong AI

Weak AI	Strong AI
Weak AI is simply the view that intelligent behaviour can be modeled and used by computers to solve complex problems.	Strong AI refers to a hypothetical machine that exhibits human cognitive abilities,
Weak AI refers to systems that are programmed to accomplish a wide range problems but operate with a pre-defined range of functions.	Strong AI refers to machines with the mind of their own and which can think and accomplish complex tasks on their own.
Weak AI-powered machines do not have ming of their own.	Strong AI-powered machines can exhibit strong human cognitive abilities.
Alexa and Siri are the best examples of weak AI programs.	Strong AI is a hypothetical concept which does not exist yet in its true

What is Intelligent System?

- Intelligence is an **experience** that one gets by interacting with system
- Intelligence is **intangible** like other attributes - fast, secure, usable, intuitive
- Intelligent system is **non-deterministic**
- Intelligence is adding **value** to businesses
- A System **appears** to be intelligent
- Are Personal Computers or typical programs (Browser etc.) intelligent?

Scope of Artificial Intelligence



Source: [What's required for a machine to be intelligent]

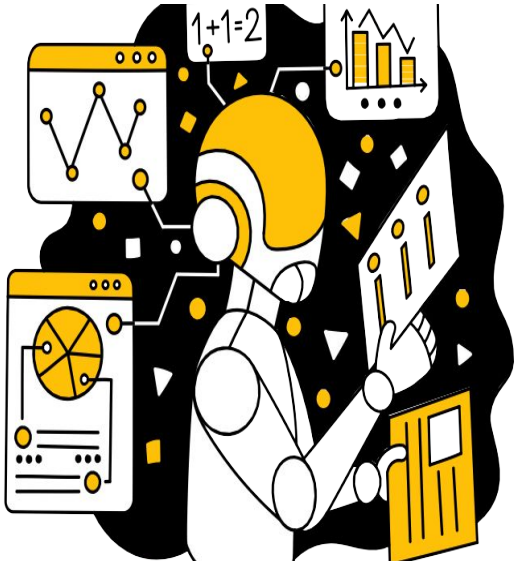
Examples of Artificial Intelligence

Speech Recognition

- Understanding and processing human

Image Recognition

- Identifying objects, faces, or patterns in images.



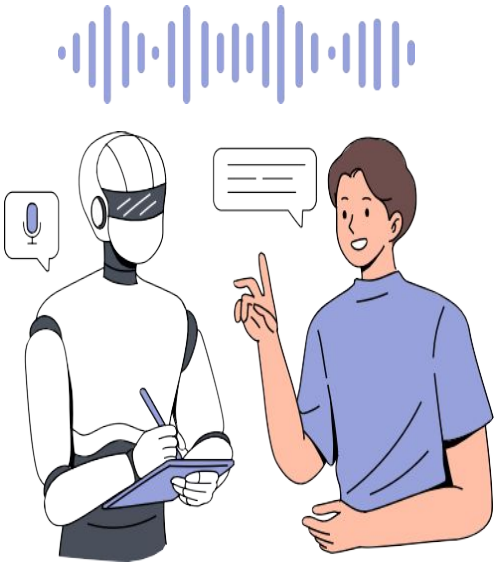
Examples of Artificial Intelligence

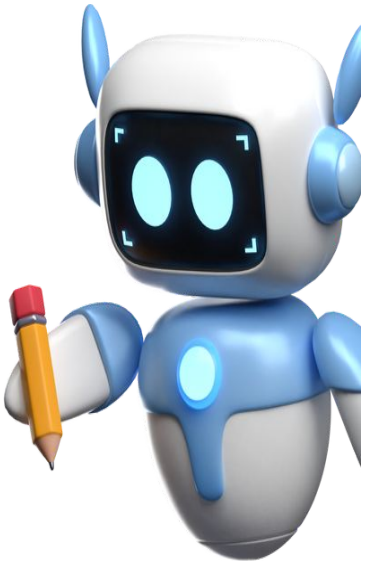
Decision Making

Analyzing data to make informed choices.

Learning New Skills

Problem-solving in complex, abstract domains.





Key Topics in Artificial Intelligence

Core AI Techniques

- **Heuristic Search:** Exploring possible solutions using intelligent trial and error.
- **Logic Representation:** Using formal logic to represent knowledge and reason about problems.
- **Symbolic Processing:** Manipulating symbols to represent real-world entities and concepts.
- **Numeric Processing:** Handling numerical data for computations and analyses.
- **Problem Solving:** Techniques for finding solutions to complex issues.

Key Topics in Artificial Intelligence

Advanced AI Methods

- **Natural Language Processing (NLP):** Understanding and generating human language.
- **Knowledge Representation:** Structuring information to make is usable for AI applications.
- **Expert Systems:** Emulating the decision-making ability of human experts.
- **Robotics:** Designing intelligent machines that can perform physical tasks.

Key Topics in Artificial Intelligence

Machine Learning and AI Subfields

- **Neural Networks:** Simulating human brain processes to recognize patterns and make decisions.
- **Learning:** Adapting behaviour based on past experiences.
- **Planning:** Strategically determining actions to achieve specific goals.
- **Semantic Networks:** Connecting concepts in a network to represent knowledge.
- **Clustering and Classification:** Grouping and categorizing data based on similarities.
- **Regression:** Predicting outcomes based on input data.
- **Control:** Directing the behaviour of machines or systems.

Deep Learning: Unlocking the Power of Data

As we entered the 2010s, the AI landscape was transformed by a breakthrough known as Deep Learning. If Machine Learning was like teaching a computer to learn from data, Deep Learning was like giving it a brain with billions of neurons, capable of processing vast amounts of information in ways that mimicked human learning.

Computer Vision emerged as a powerful tool that allowed machines to interpret visual data—recognizing objects, faces, and even entire scenes with incredible accuracy. Powered by advances in Deep Learning, Computer Vision transformed industries like security, healthcare, and retail.

Natural Language Processing: Teaching Machines to Understand Us

Understanding images and playing games was just the beginning. The real challenge was teaching machines to understand and communicate with us in our own language.

Natural Language Processing (NLP), a field of AI dedicated to bridging the gap between human language and machine understanding.

NLP had been around for decades, but it wasn't until the 2010s, with the advent of models like BERT and GPT, that machines started to truly grasp the nuances of language.

These models could understand context, interpret meaning, and even generate human-like text.

Suddenly, talking to a machine didn't feel so different from talking to a human, and applications like virtual assistants, chatbots, and translation services began to thrive.

Autonomous Systems: Machines on the Move

Autonomous systems, like self-driving cars and drones, represent one of the most exciting frontiers of AI. These systems combine AI with sensors, cameras, and real-time data processing to navigate the world without human intervention.

The development of autonomous systems posed unique challenges—

How do you teach a car to make split-second decisions on a busy road?

How do you ensure a drone can safely deliver a package?

By integrating AI with advanced sensors and sophisticated algorithms, engineers have created machines that can operate independently, transforming industries from transportation to logistics.

Generative AI: Unleashing the Creativity

As AI reached new heights, it began to explore the realm of creativity—a domain once thought to be uniquely human.

Generative AI models, like GPT-3, have the ability to create original content, from writing articles to composing music and generating artwork.

Generative AI has opened up a world of possibilities in art, design, entertainment, and beyond. It's like having a creative partner that can brainstorm ideas, draft content, or even paint a picture.

But with this power comes responsibility—ensuring that AI-generated content is used ethically and transparently is a challenge that the AI community continues to tackle.

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Introduction

- Artificial intelligence is the simulation of human intelligence processes by developing computing systems that have the ability to perform tasks like human.
- AI is the study of ideas that enable computers to be intelligent.
- Applications of AI include Expert System, Natural Language Processing, Speech Recognition, Robotics and Machine Vision.
- AI can process large amounts of data much faster and make predictions more accurately than humanly possible.
- AI is expensive to process the large amounts of data.

What is *artificial* intelligence?

human-like *vs.* rational

thought *vs.* behavior		
	Systems that think like humans	Systems that think rationally
	Systems that act like humans	Systems that act rationally

- Cognitive Science
  - Very hard to understand how humans think

We don't do cost benefit analysis, rather decision is intuitive

- Thinking like humans important in Cognitive Science applications
  - Intelligent tutoring
  - Expressing emotions in interfaces… HCI

Systems that think ‘rationally’

- Humans are not always ‘rational’
- How to define *Rationale* - Rationale means doing the right thing
- *The right thing* is expected to maximize goal achievement, given the available information
- Is it in terms of *logic*?
- Logic can’t express everything (e.g. uncertainty)
- At that time, some *specific (in domain) human knowledge* or information is used, thus covers more generally different situations of problems.
- AI is the study of mental facilities through the use of computational models (Charniak and McDermott).
- “The study of the computations that make it possible to perceive, reason, and act” (Winston)

Systems that act Rationally

- Study AI as *rational* ***agent*** –

<u>2 advantages:</u>

  - It is more general than using logic only
    - Because: LOGIC + Domain knowledge
  - It allows extension of the approach with more scientific methodologies

AI has two major roles:

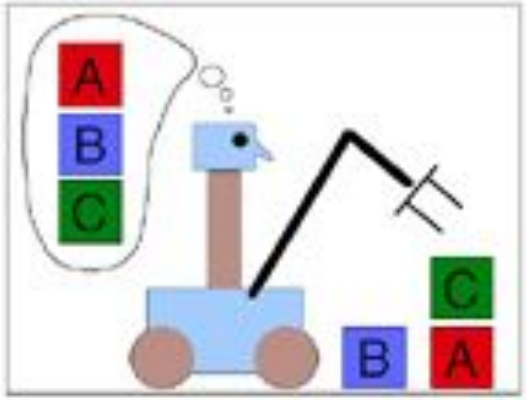
  - Study the intelligent part concerned with humans.
  - Represent those actions using computers.

Systems that act like humans

- *Natural language processing*
  - for communication with human
- *Knowledge representation*
  - to store information effectively & efficiently
- *Automated reasoning*
  - to retrieve & answer questions using the stored information
- *Machine learning*
  - to adapt to new circumstances

Goals of AI

- To make computers more useful by letting them take over dangerous or tedious tasks from human.
- Understand principles of human intelligence



The Foundation of AI

- Mathematics formalizes the three main area of AI: *computation*, *logic*, and *probability*
  - Computation leads to analysis of the problems that can be computed
    - *complexity theory*
  - Probability contributes the *"degree of belief"* to handle *uncertainty* in AI
  - *Decision theory* combines *probability theory* and *utility theory* (bias)

- *Psychology*
  - How do humans think and act?
  - The study of human reasoning and acting
  - Provides reasoning models for AI
  - Strengthen the ideas
    - humans and other animals can be considered as information processing machines

The Foundation of AI

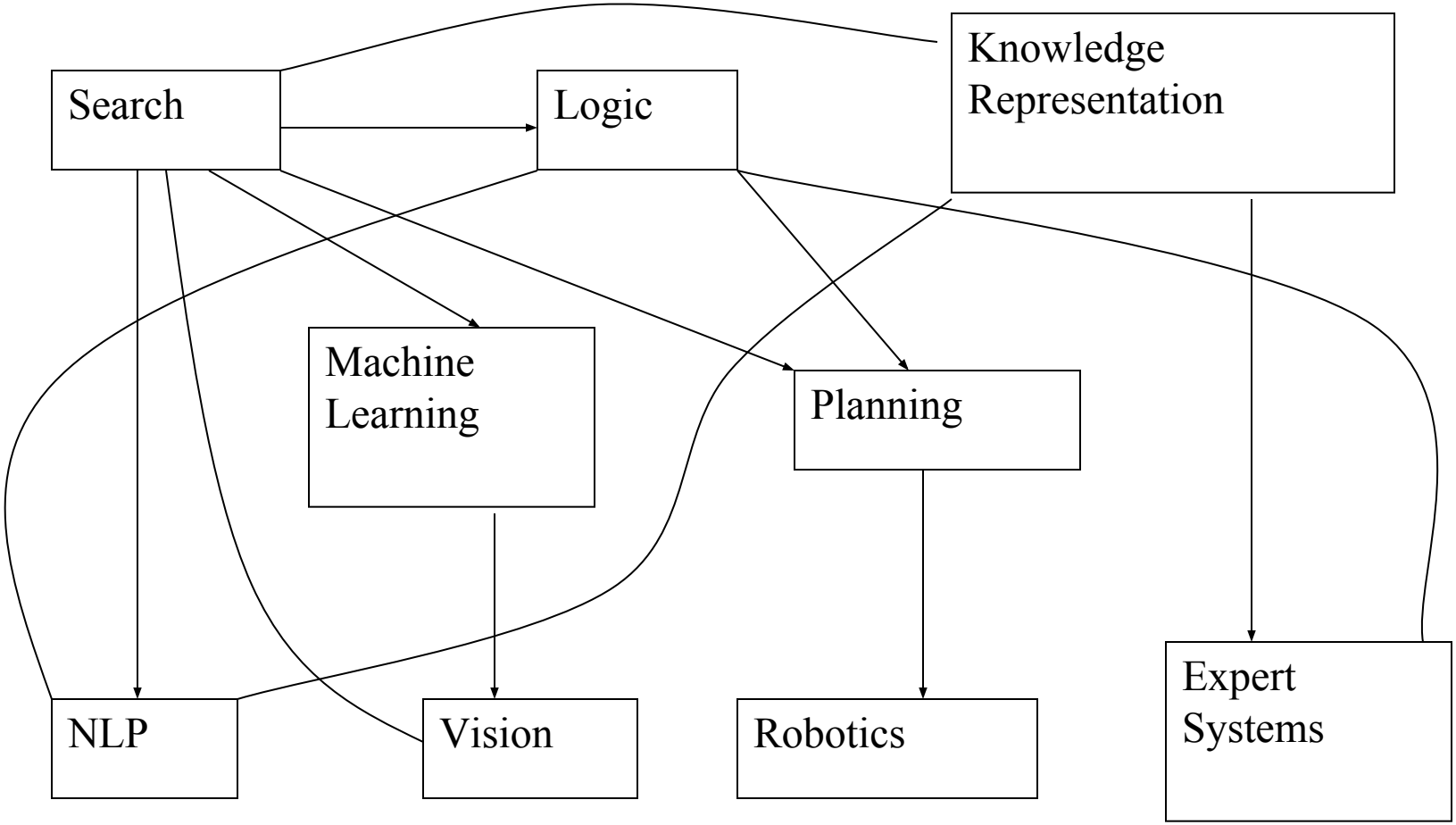
- Computer Engineering
  - The power of computer makes computation of large and difficult problems more easily
  - AI has also contributed its own work to computer science, including: time-sharing, the linked list data type, OOP, etc.
- Linguistics
  - For understanding natural languages
    - different approaches has been adopted from the linguistic work
  - Formal languages
  - Syntactic and semantic analysis
  - Knowledge representation

The main topics in AI

Artificial intelligence can be considered under a number of headings:

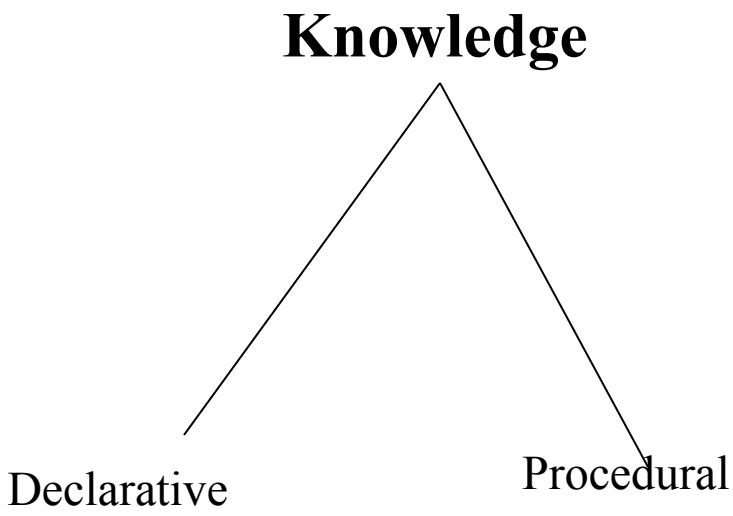
- Search (includes Game Playing).
- Representing Knowledge and Reasoning with it.
- Planning.
- Learning.
- Natural language processing.
- Expert Systems.
- Interacting with the Environment
  (e.g. Vision, Speech recognition, Robotics)

Areas of AI and Some Dependencies



Search

- *Search* is <u>the fundamental</u> technique of AI.
  - Possible answers, decisions or courses of action are structured into an abstract space, which we then search.
- Search is "blind" and “informed":
  - blind
    - we move through the space without worrying about what is coming next, but recognising the answer if we see it
  - informed
    - we guess what is ahead, and use that information to decide where to look next.
- We may want to search for the first answer that satisfies our goal, or we may want to keep searching until we find the best answer.



- Declarative knowledge deals with factoid questions (what is the capital of India? Etc.)
- Procedural knowledge deals with “How”
- Procedural knowledge can be embedded in declarative knowledge

Planning

Given a set of goals, construct a sequence of actions that achieves those goals:

- often very large search space
- but most parts of the world are independent of most other parts
- often start with goals and connect them to actions
- no necessary connection between order of planning and order of execution
- what happens if the world changes as we execute the plan and/or our actions don't produce the expected results?

AI Planning

Elements of a Planning Problem

- A set of states (worlds) described in terms of predicates
- A set of actions which transforms some parts of one world to take us to another world
- An initial world
- A goal in terms of the predicates that must hold in the final world

Planning is widely used in robotics and automated control

Modern AI explores techniques that combine planning with machine learning

Learning

- If a system is going to act truly appropriately, then it must be able to change its actions in the light of experience:
  - how do we generate new facts from old ?
  - how do we generate new concepts ?
  - how do we learn to distinguish different situations in new environments ?

Interacting with the Environment

- In order to enable intelligent behaviour, we will have to interact with our environment.
- Properly intelligent systems may be expected to:
  - accept sensory input
    - vision, sound, …
  - interact with humans
    - understand language, recognise speech, generate text, speech and graphics, …
  - modify the environment
    - robotics

Knowledge and Inferences

- Several ways to know:
  - **Empiricism:** Involves acquiring knowledge through observations or experiences.
  - The **Scientific** method: A process of systematically collecting and evaluating evidence to test ideas and answer questions.
  - **Intuition**s: Rather than examining facts or using rational thoughts, intuition involves believing what feels true.
  - **Authority**: Accepting ideas because some authority figure says that they are true.
  - **Rationalism**: Using Logic and Reasoning to acquire knowledge

Reasoning = Making Inference

Logical Reasoning

- What is known to reason about something that is not known explicitly.
- **Deduction** : From a given set of facts infer another fact that is *necessarily* true – *cause to effect* (disease to symptoms)
- **Abduction**: From a given set of facts infer another fact that is *possibly* true – *Effect to cause* (symptoms to disease).
  - Abduction needs to be consistent.
  - Abduction is the guess work.
- **Induction**: From a given set of facts infer a new fact.
  - Known as generalization.
  - Recognizing that a number of entities in the domain share some common property, and assert as a general statement.

Automated Reasoning

- **Deduction**

  Rule: *All the marbles in this bag are blue*
  Case: *These marbles are from this bag*
  Inference: *These marbles are blue*

- **Abduction**

  Rule: *All the marbles in this bag are blue*
  Observation: *These marbles are blue*
  Case: *These marbles are from this bag*

- **Induction**

  Case: *These marbles are from this bag*
  Observation: *These marbles are blue*

  Rule: All the marbles in this bag are blue

Logical Reasoning

father(x, z), father(z, y)
⇒ grandfather(x, y)

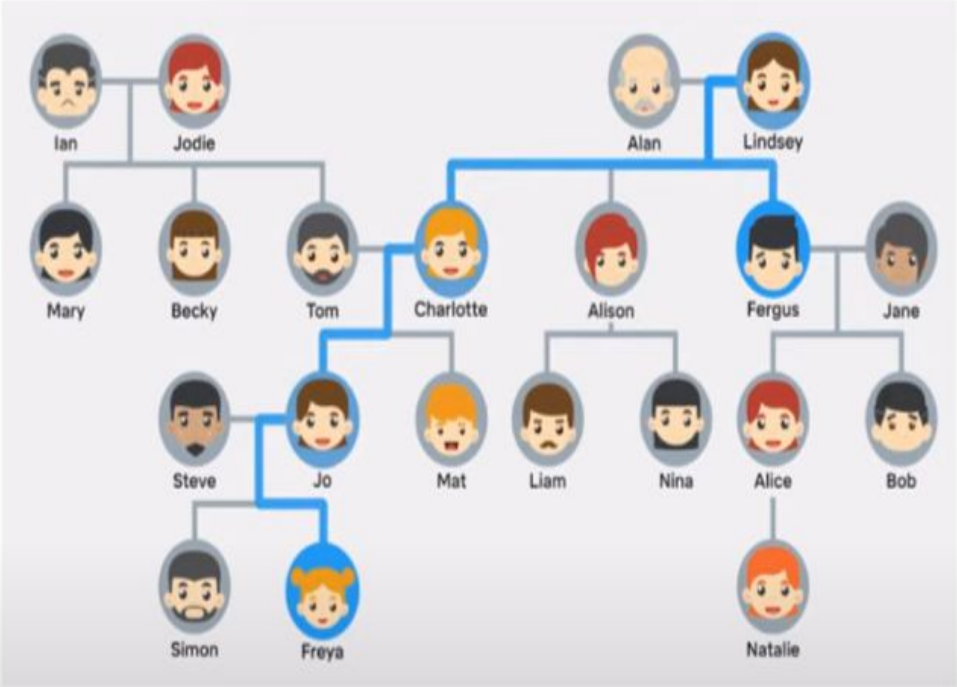
mother(x, z), father(z, y)
⇒ maternalgrandfather(x, y)

mother(x, z), mother(z, y)
⇒ maternalgrandmother(x, y)

father(x, z), mother(z, y)
⇒ grandmother(x, y)

maternalgrandmother(x, z),
mother(z, p), son(p, y)
⇒ maternalgreatuncle(x, y)

Who is the maternal great uncle of Freya?



maternalgrandmother(Freya, Charlotte),
mother(Charlotte, Lindsey), son(Lindsey, Fergus)
⇒ maternalgreatuncle(Freya, Fergus)

Problem Solving

- An autonomous agent has a goal to achieve by choosing a set of actions.
- Abductive inference suggests what action is best.
- Background knowledge make some strategies and tactics where knowledge comes through training.
- Abductive inferences generate **expectations**.
- Scientific progress begins with **Abduction** (looking for causes) and **Induction** (making informed Generalization).
- Gets validated by Deduction
- Gets experimental Validation
- Only Induction adds New Knowledge.

Constraint optimization problem

When the size and complexity becomes too big we use "heuristic functions" to cut out unnecessary parts.

In the lack of domain knowledge, we can statistically learn the best way (reinforcement learning) by exploration.

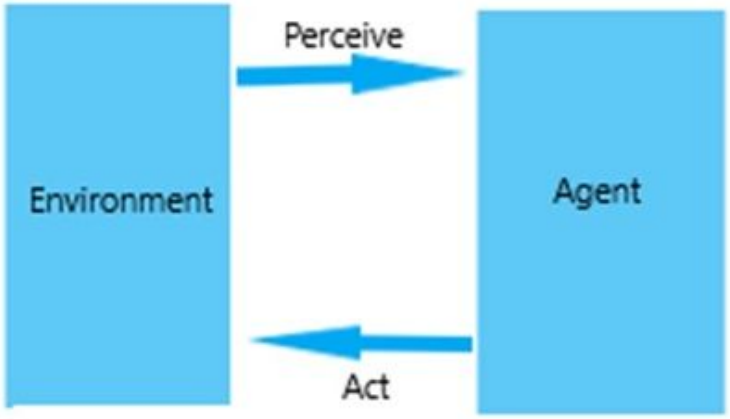
Modern AI aims to combine learning from data with structured use of domain knowledge.

Agent and Environment

- **Artificial intelligence** studies the whole system by dividing it into two components:

  Agent

  Environment
- An agent can either perceive information from the environment or can perform some actions on the environment.



Agent

- The system which is responsible for acting on the environment and causing any changes to it is called an agent.
- It first perceives the information by observing the environment and then acts according to the particular scenario.
- An agent is the one which should be designed in such a way that it may interact from its surroundings by itself, i.e. it should not require any user input to feed information into it.
- It should be able to understand the situation, find the best decision that could be taken, and then act according to it.

The objective of AI is to explore the ways onto a machine that can:

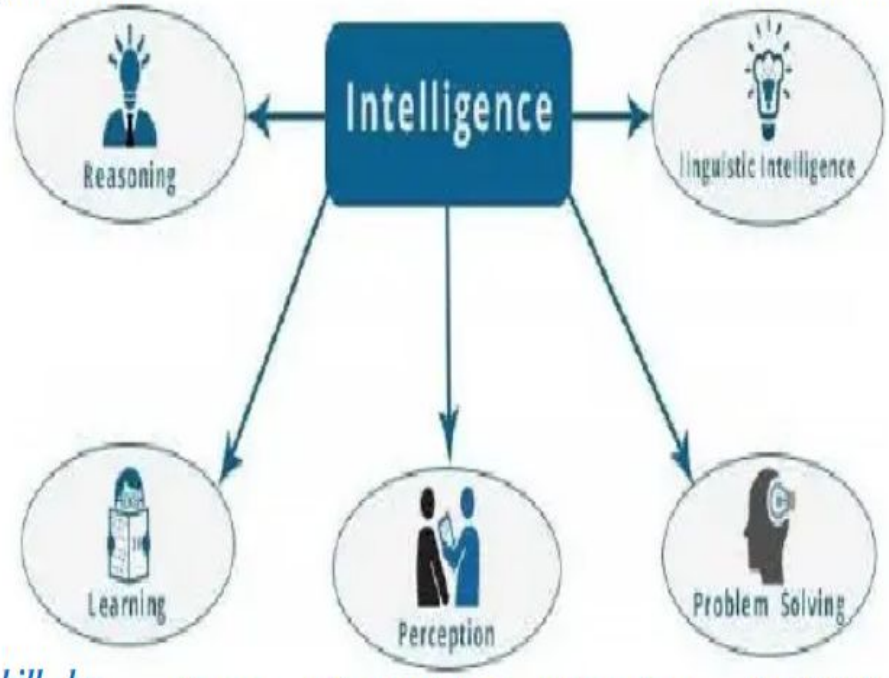
- **Reason like a human**,
- **Think like a human**
- **Act like a human**.

- Its approach is to train a machine (i.e., a computer or a robot) with the **same capabilities as of a human brain.**

- The intelligence is **intangible.**

Set of processes for judgement, making decisions, and prediction

Ability to use, comprehend, speak, and write the verbal and written language



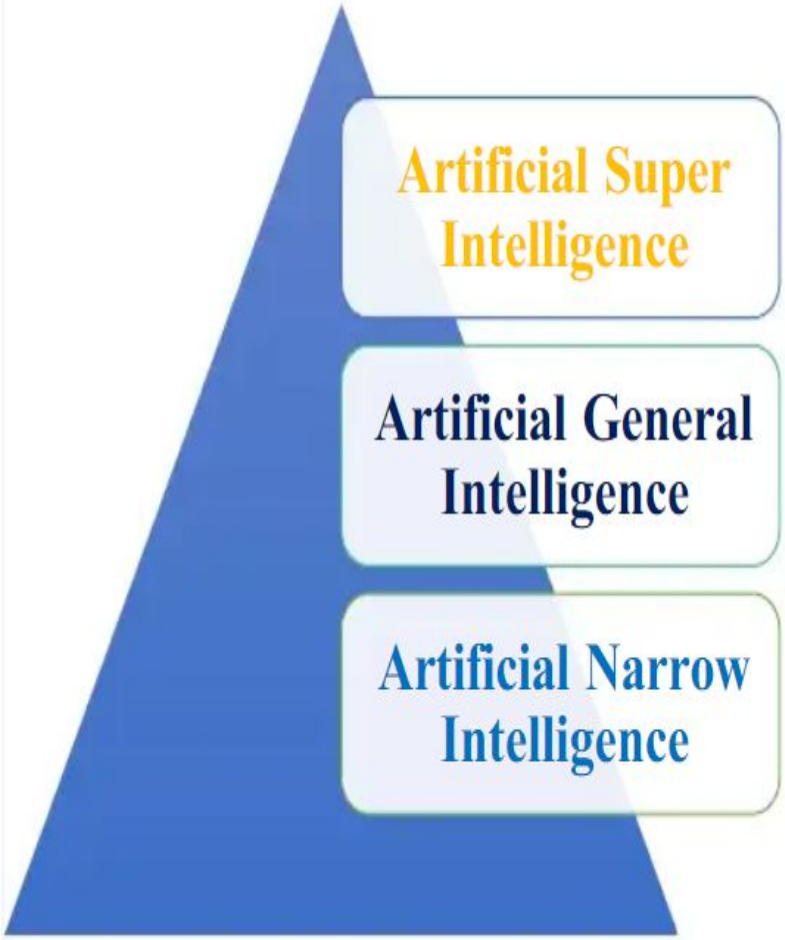
Gaining knowledge or skill by studying, practicing, being taught.

Process of acquiring, interpreting, selecting, and organizing sensory information.

Process in which one perceives and tries to arrive at a desired solution

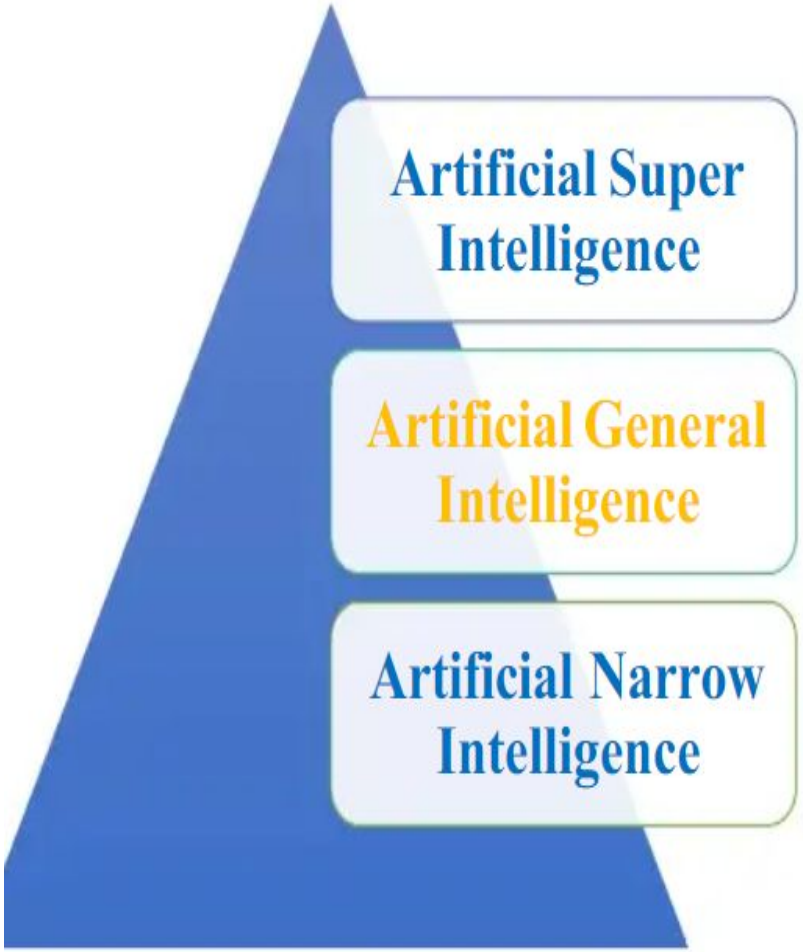
- Consider a scenario where a **human brain may fail** to take **an intelligent decision** and **need someone** who can make a **wise and intelligent decision** for him.
- AI can create such **software or devices** which can solve **real-world problems** very easily and with accuracy such as **health issues**, **marketing**, **traffic issues**, etc.
- AI can create your personal **VA's**, such as **Cortana**, **Google Assistant**, **Siri**, etc.
- AI can build such Robots which can work in an environment where **survival of humans** can be at **risk.**

Stages Of Artificial Intelligence



Artificial Narrow Intelligence (ANI)

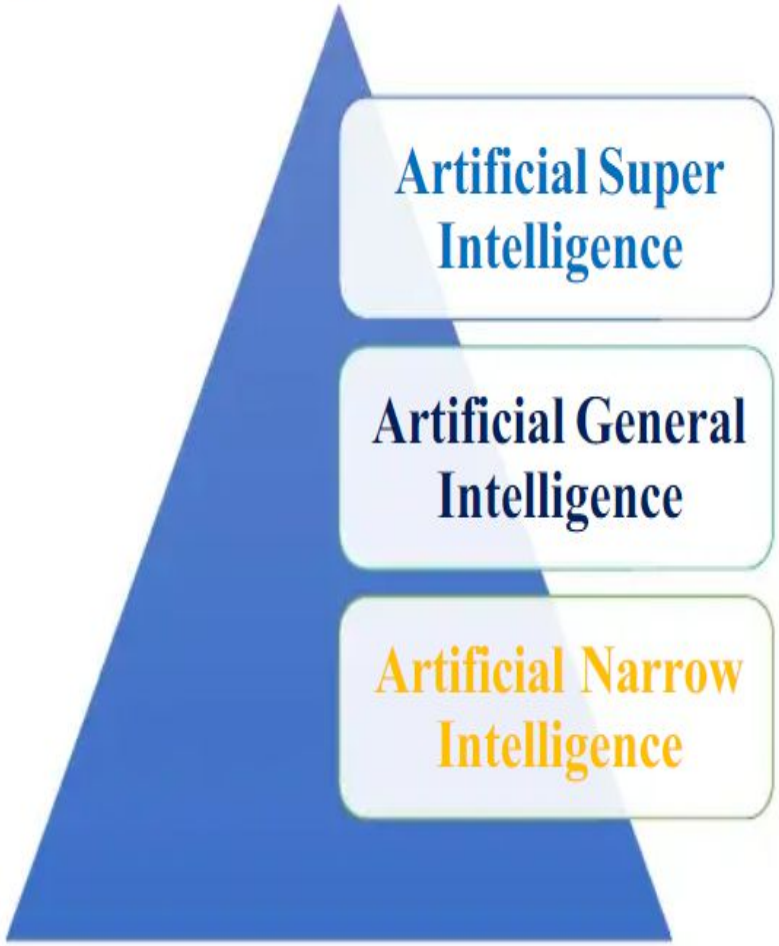
- Also known as **Weak AI,** involve machines that can **perform only narrowly** defined **set of specific tasks**.
- The machine **does not possess any thinking ability**, just performs **a set of pre-defined functions.**
- ***Siri, Alexa, Self-driving cars, Alpha-Go*** based on Weak AI.



Artificial General Intelligence (AGI)

- Also known as **Strong AI**, AGI is the stage of **evolution** of Artificial Intelligence.
- These machines **will possess** the **ability to think and make decisions** just like us humans.
- There are currently no existing examples of **Strong AI**, however, it is believed that we will soon be able to create machines that are **as smart as humans.**

Stages Of Artificial Intelligence



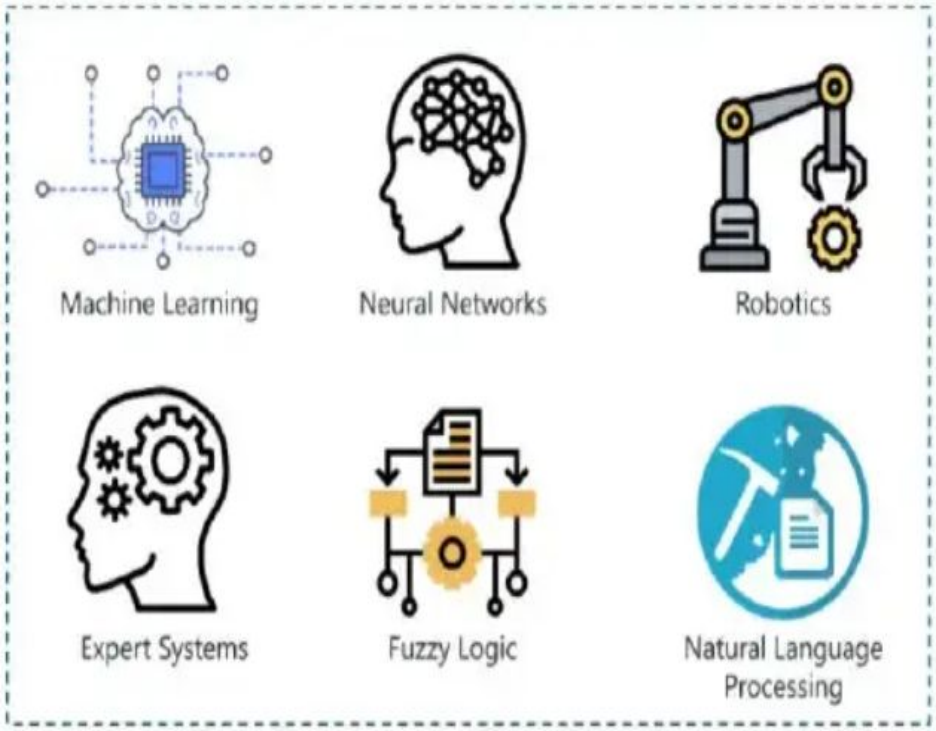
Artificial Super Intelligence (ASI)

- Artificial Super Intelligence is the **stage of Artificial Intelligence** when **the capability of computers** will ***surpass human beings***.
- ASI is currently a **hypothetical situation** as **depicted in movies** and science fiction books, where ***machines have taken over the world***.

Branches Of Artificial Intelligence

Artificial Intelligence can be used to **solve real-world problems** by **implementing the following processes/ techniques:**

- **Machine Learning**
- **Deep Learning**
- **Natural Language Processing**
- **Robotics**
- **Expert Systems**
- **Fuzzy Logic**



Major Use Cases Of Artificial Intelligence

- **Artificial Intelligence In Sports**

Deep Blue was a chess-playing computer developed by IBM. It is the **first computer chess-playing system** to win both a chess game and a chess match against a leading **world champion**.

- **Artificial Intelligence For Rescue Missions**

The use of Artificial Intelligence and technology to ensure the **help arrives faster**. Developing systems which help **first responders** find **victims of earthquakes, floods**, and any **other natural disasters** like **Covid-19.**

- **Artificial Intelligence For Smart Agriculture**

Complete monitoring of the soil and crop yield to providing **predictive analytic models** to track and predict various factors and variables that could **affect future yields**.

The Berlin-based agricultural tech startup **PEAT** has developed a **deep learning algorithm-based application** called **Plantix** which can identify **defects and nutrient deficiencies in the soil.**

- **Artificial Intelligence In Healthcare**

These machines **enable shortening the patients' hospital stay**, **positively affecting the surgical experience** and **reducing medical costs** all at once.

Similarly, **mind-controlled robotic arms and brain chip implants** have begun helping **paralyzed patients regain mobility and sensations of touch**.

- **Artificial Intelligence Tracking Wildlife Populations**

Applications like **iNaturalist** and **eBirds** collect data on the **species encountered**. This helps keep **track of species populations, ecosystems and migration patterns**.

As a result, these applications also have an **important role** in the better identification and **protection of marine and freshwater ecosystems** as well.

Summary of AI Techniques

AI techniques are knowledge-based methods designed to solve complex problems.

Knowledge in AI must be well-structured, adaptable, and provide generalizations.

The goal is to make AI techniques flexible and usable across various problem domains.

AI techniques are problem independent.

In order to try to characterize AI techniques in as problem-independent a way as possible, let's look at two very different problems and a series of approaches for solving each of them.